ALGEBRA 2 · UNIT 6: EXPONENTIAL & LOGARITHMIC FUNCTIONS

Earthquake & Sound Lab

Use logarithms and exponential models to compare earthquakes, sounds, and ancient bones.

CP1

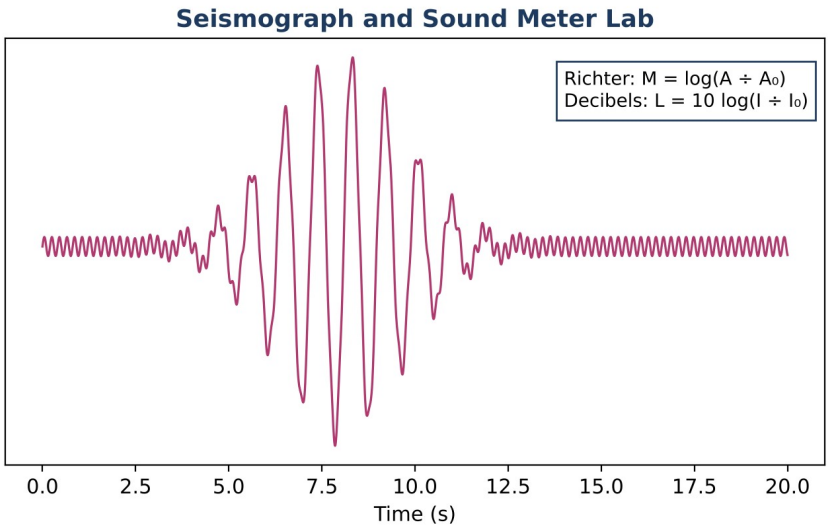
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Scientists measure huge ranges of values: tiny tremors and enormous earthquakes, whispers and jet engines. Logarithmic scales squeeze these ranges into small numbers.

You will compare quakes and sounds, date ancient bones with carbon-14, model cooling, and present what you learn on a safety poster.



Learning Goals

- I can solve exponential equations using logarithms.
- I can use log properties to rewrite expressions.
- I can model with exponential decay and Newton's law of cooling.

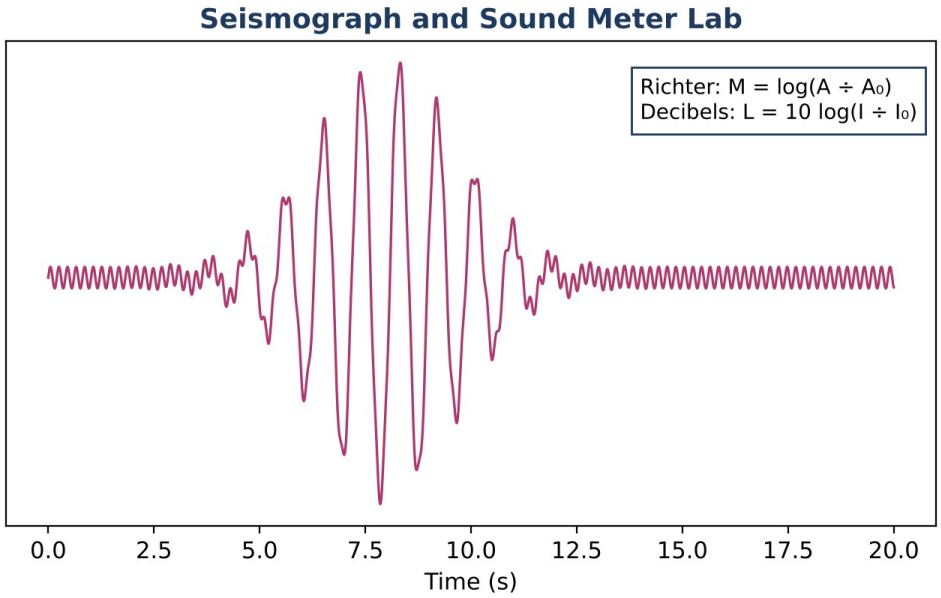
Project Timeline

When	What you will do
Day 1	Log Scales
Day 2	Solve with Logs
Day 3	Safety Poster
Final	Science Safety Poster + reflection

Part 1: Log Scales

Day 1 · Use the formulas on the hero image.

1. Compare the amplitudes of magnitude 6.8 and 5.3 quakes.



2. Each increase of 1 in magnitude multiplies the energy by about $10^{1.5} \approx 31.6$. About how many times more energy does a magnitude 7 release than a magnitude 5?

3. Condense $\log 2 + \log 50$ and expand $\log(100x^3)$.

Part 2: Solve with Logs

Day 2 · Carbon-14: $A = A_0(0.5)^n$, where $n = t \div 5730$ is the number of half-lives.

4. A fossil has 30% of its carbon-14 left. Solve for its age with logarithms.

5. Coffee cools by $T = 70 + 130e^{-0.08t}$ (°F, minutes). When will it reach 100°F?

6. Solve $2^{x+3} = 50$.

Part 3: Safety Poster

Day 3 · Create a poster with log comparisons and one model.

7. Compare three events with log scales.

8. Include one exponential model solved with logs.

9. Write a safety message for the public.

Final Product: Science Safety Poster

Your final product must include:

- Log-scale comparisons
- Solved exponential equations
- Poster

Reflection

R1. How do logarithms undo exponents?

R2. Why does the coffee never cool below 70°F in the model?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Comparisons and models are correct and explained in plain language for the public. Goes beyond the minimum with a creative, realistic choice.	Comparisons and models are correct and explained in plain language for the public.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Poster shows the log scales, at least three comparisons with work, and an exponential model solved with logarithms. Polished, organized, and visually engaging.	Poster shows the log scales, at least three comparisons with work, and an exponential model solved with logarithms. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.